

Assignment 2

Virtualisation and cloud computing

Recorded Video Demo: [demo](#)

Viswanadhapalli Sujay
B22CS063

1. Introduction

This report outlines the step-by-step implementation of a scalable and secure Virtual Machine (VM) setup on Google Cloud Platform (GCP). It covers the creation of VM instances, the configuration of auto-scaling policies, and security enhancements such as IAM role management and firewall rule configurations. The choices made in this setup were based on practical testing and best practices for scalability and security.

2. VM Instance Setup

Step 1: Accessing GCP Console

I logged into Google Cloud Console and navigated to **Compute Engine > VM Instances** to manage my cloud resources.

Step 2: Creating the VM Instance

To ensure smooth scalability, I deployed an initial instance with a balanced configuration:

- **Instance Name:** `my-instance`
- **Zone:** `us-central1-c` (chosen for lower latency and cost-efficiency)
- **Machine Type:** `e2-medium` (2 vCPUs, 4 GB RAM) – selected to balance cost and performance.
- **Boot Disk:** Debian 12 (lightweight and stable for web server and application hosting)
- **Firewall Rules:** Enabled **HTTP** and **HTTPS** to allow web traffic.

I accessed the VM via SSH:

Bash:

```
gcloud compute ssh my-instance --zone=us-central1-c
```

Machine configuration

Machine type	e2-medium (2 vCPUs, 4 GB Memory)
CPU platform	Intel Broadwell
Minimum CPU platform	None
Architecture	x86/64
vCPUs to core ratio ?	—
Custom visible cores ?	—
All-core turbo-only mode ?	—
Display device	Disabled Enable to use screen capturing and recording tools
GPUs	None
Resource policies	

Firewalls

HTTP traffic	On
HTTPS traffic	On
Allow Load Balancer Health checks	On

Network tags

http-serverhttps-serverlb-health-check

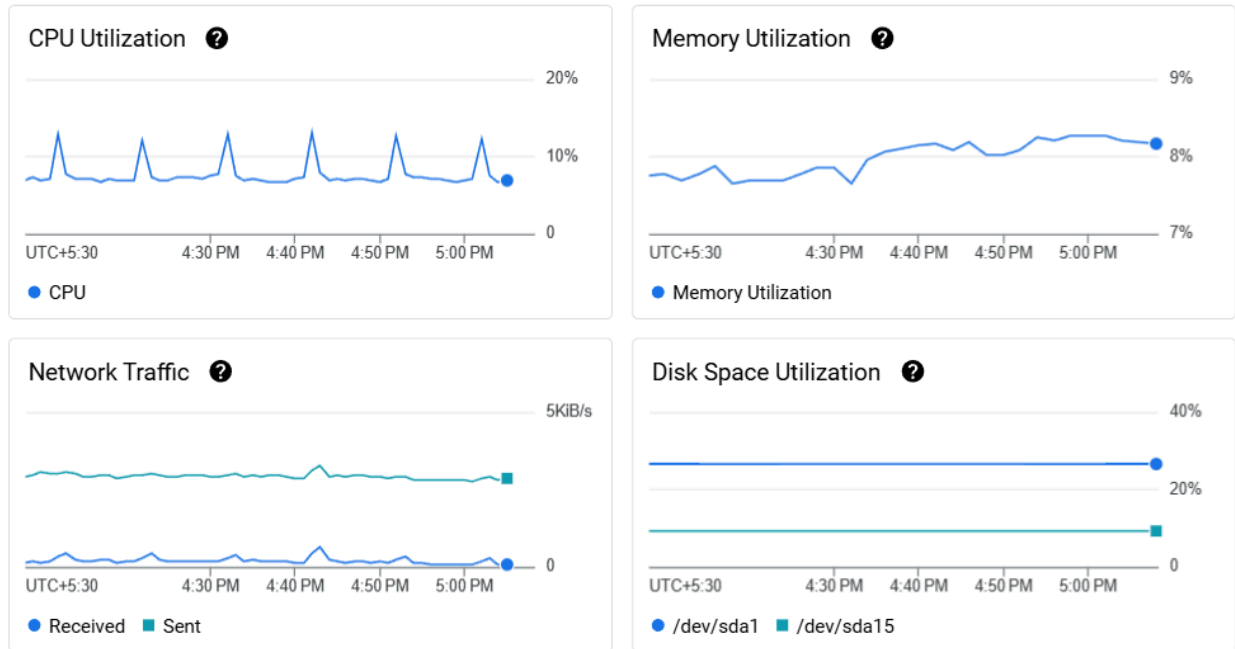
Network interfaces

Name ↑	Network	Subnetwork	Primary internal IP address	Alias IP ranges	IP stack type	External IP address ↑	Network
nic0	default	default	10.128.0.2		IPv4	34.122.190.6 (Ephemeral)	Public

Storage

Boot disk

Name ↑	Image	Interface type	Size (GB)	Device name	Type	Architecture	Encryption	Mode	Wh
my-instance	debian-12-bookworm-v20250212	SCSI	10	my-instance	Balanced persistent disk	x86/64	Google-managed	Boot, read/write	Del



3. Configuring Auto-Scaling

To ensure the system scales based on demand, I created an **Instance Group** with auto-scaling policies.

Step 1: Creating a Managed Instance Group

I navigated to **Compute Engine > Instance Groups** and configured:

- **Instance Group Name:** `instance-group-1`
- **Base Instance Name:** `instance-group-1-qpnr` (assigned by GCP)
- **Zone:** `us-central1-c`
- **Instance Template:** A template matching `my-instance` configuration.

Basic information

Name	instance-template-20250302-101331
Type	Instance Template
Creation time	Mar 2, 2025, 3:45:03 PM UTC+05:30
In use by	instance-group-1
Location	us-central1
Reservations	Automatically choose
Labels	None
Tags	—
Placement policy	No policy
Confidential VM service	Disabled

Machine configuration

Machine type	e2-medium
Minimum CPU platform	None
Architecture	—
vCPUs to core ratio	—
Custom visible cores	—
All-core turbo-only mode	—
Display device	Disabled
GPUs	None

Storage

Boot disk

Name ↑	Image	Interface type	Size (GB)	Device name	Type	Architecture	Encryption	Mode
Autogenerated	debian-12-bookworm-v20250212	—	10	instance-template-20250302-101331	Balanced persistent disk	—	Google-managed	Boot, read/write

- **Autoscaling Mode:** Enabled
- **Scaling Policy:** Based on CPU utilization
- **Target CPU Utilization:** 60% (Instances will scale up if CPU usage exceeds this threshold)
- **Minimum Instances:** 1
- **Maximum Instances:** 5 (chosen after testing to balance cost and performance)

I selected **CPU utilization** as the scaling metric since it's the most reliable indicator of system load.

Autoscaling

Autoscaling mode	On
Minimum # of instances	1
Maximum # of instances	5
Initialization period	60 seconds
Autoscaling signal	
CPU utilization	60%
Predictive autoscaling	Off
Scale in controls	Off
Scaling schedules	MANAGE SCHEDULES

Instance groups are collections of VM instances that use load balancing and automated services, like autoscaling and autohealing. [Learn more](#)

Filter Enter property name or value								
☐ Status	Name ↑	Instances	Template	Group type	Creation time	Recommendation	Autoscaling	Zone
☒	instance-group-1	1	instance-template-20250302-101331 (Regional)	Managed	Mar 2, 2025, 3:48:00 PM UTC+05:30		On: Target CPU utilization 60%	us-central1-c

Step 2: Stress Testing Auto-Scaling

To test the auto-scaling behavior, I simulated high CPU usage:

bash:

```
sudo apt update && sudo apt install stress -y
stress --cpu 8 --timeout 300
```

And top command

```
SSH-in-browser

top - 10:58:17 up 39 min, 3 users, load average: 6.37, 3.08, 2.01
Tasks: 99 total, 9 running, 90 sleeping, 0 stopped, 0 zombie
%Cpu(s): 99.2 us, 0.5 sy, 0.0 ni, 0.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.3 st
MiB Mem : 3924.7 total, 3163.3 free, 419.1 used, 569.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3505.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
 1140 b22cs063  20   0   3364    108     0  R   25.0   0.0   0:23.31 stress
 1142 b22cs063  20   0   3364    108     0  R   25.0   0.0   0:23.35 stress
 1144 b22cs063  20   0   3364    108     0  R   25.0   0.0   0:23.42 stress
 1146 b22cs063  20   0   3364    108     0  R   25.0   0.0   0:23.41 stress
 1141 b22cs063  20   0   3364    108     0  R   24.7   0.0   0:23.25 stress
 1143 b22cs063  20   0   3364    108     0  R   24.7   0.0   0:23.32 stress
 1145 b22cs063  20   0   3364    108     0  R   24.7   0.0   0:23.19 stress
 1147 b22cs063  20   0   3364    108     0  R   24.7   0.0   0:23.34 stress
   800 root      20   0 1241356 22836 14568  S    1.3   0.6   0:55.82 google_guest_ag
     1 root      20   0  102328  12216   9120  S    0.0   0.3   0:01.80 systemd
     2 root      20   0     0     0     0  S    0.0   0.0   0:00.00 kthreadd
     3 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 rcu_gp
     4 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 rcu_par_gp
     5 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 slub_flushwq
     6 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 netns
     8 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    10 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 mm_percpu_wq
    11 root     20  0     0     0     0  I    0.0   0.0   0:00.00 rcu_tasks_kthread
    12 root     20  0     0     0     0  I    0.0   0.0   0:00.00 rcu_tasks_rude_kthread
    13 root     20  0     0     0     0  I    0.0   0.0   0:00.00 rcu_tasks_trace_kthread
    14 root     20  0     0     0     0  S    0.0   0.0   0:00.03 ksoftirqd/0
    15 root     20  0     0     0     0  I    0.0   0.0   0:00.07 rcu_preempt
    16 root      rt  0     0     0     0  S    0.0   0.0   0:00.00 migration/0
    18 root     20  0     0     0     0  S    0.0   0.0   0:00.00 cpuhp/0
    19 root     20  0     0     0     0  S    0.0   0.0   0:00.00 cpuhp/1
    20 root      rt  0     0     0     0  S    0.0   0.0   0:00.46 migration/1
    21 root     20  0     0     0     0  S    0.0   0.0   0:00.03 ksoftirqd/1
    22 root     20  0     0     0     0  I    0.0   0.0   0:00.07 kworker/1:0-events
    23 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 kworker/1:0H-events_highpri
    26 root     20  0     0     0     0  S    0.0   0.0   0:00.00 kdevtmpfs
    27 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 inet_frag_wq
    28 root     20  0     0     0     0  S    0.0   0.0   0:00.02 kauditd
    29 root     20  0     0     0     0  S    0.0   0.0   0:00.00 khungtaskd
    31 root     20  0     0     0     0  S    0.0   0.0   0:00.00 oom_reaper
    32 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 writeback
    33 root     20  0     0     0     0  S    0.0   0.0   0:00.06 kcompactd0
    34 root     25  5     0     0     0  S    0.0   0.0   0:00.00 ksmd
    35 root     39 19     0     0     0  S    0.0   0.0   0:00.05 khugepaged
    36 root      0 -20     0     0     0  I    0.0   0.0   0:00.00 kintegrityd
```

I monitored the instance group in **GCP Console > Instance Groups** to see new instances launching.

VM instances

 Filter Enter property name or value

☐ Status	Name 	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐ 	instance-group-1-1k7g	us-central1-c		instance-group-1	10.128.0.9 (nic0)	35.193.137.151 (nic0)	SSH  
☐ 	instance-group-1-cswg	us-central1-c		instance-group-1	10.128.0.8 (nic0)	34.60.221.243 (nic0)	SSH  
☐ 	instance-group-1-dll7	us-central1-c		instance-group-1	10.128.0.10 (nic0)	35.225.177.202 (nic0)	SSH  
☐ 	instance-group-1-n3wc	us-central1-c		instance-group-1	10.128.0.11 (nic0)	34.58.30.205 (nic0)	SSH  
☐ 	instance-group-1-qpnr	us-central1-c		instance-group-1	10.128.0.5 (nic0)	34.42.193.100 (nic0)	SSH  

☰ Filter Enter property name or value

☐	Status	Name ↑	Instances
☐	✓	<u>instance-group-1</u>	5

Observation:

- The system successfully added instances when CPU usage exceeded 60%.
- When load reduced, instances were automatically removed.

Scaling efficiency was confirmed by checking:

bash:

```
gcloud compute instance-groups list-instances instance-group-1
```

4.1 Configuring IAM Roles and Firewall Rules for Secure Access

To ensure both **secure access control** and **proper network configuration**, I configured IAM roles for user-based access control and set up firewall rules to allow necessary traffic while maintaining security.

Step1 :Configuring IAM Roles for Secure Access

IAM roles are crucial for enforcing **least privilege access**, preventing unauthorized modifications, and ensuring proper system monitoring.

1. Open the **Google Cloud Console**.
2. Navigate to **IAM & Admin** → **IAM**.
3. Click **+ Add** to grant access to a user.
4. **Enter the user's email address** and select the appropriate role:
 - **Compute Viewer** → Read-only access to view instances. (*Assigned to my friend for monitoring purposes.*)
 - **Compute Admin** → Full control over VMs (Not assigned to prevent accidental modifications).
 - **Editor** → Modify instances but not billing (Not required for this scenario).
5. Click **Save** to apply the role.

Step 2: Testing IAM Role Assignment

To verify the role assignment, I asked my friend to log in and confirm that they could **view instances but not modify them**. The test was successful, confirming the IAM setup was correct.

4.2 Configuring Firewall Rules for Secure Network Access

Firewall rules are essential for **controlling inbound and outbound traffic** to VM instances, ensuring that only necessary connections are allowed.

Steps to Configure Firewall Rules

1. Open **Google Cloud Console**.
2. Navigate to **VPC Network → Firewall**.
3. Click **Create Firewall Rule**.
4. **Set Firewall Rule Details:**
 - **Name:** `allow-http-ssh`
 - **Network:** `default`
 - **Direction:** `Ingress` (Incoming traffic)
 - **Target:** `All instances in the network` (*Ensures uniform rule application to all VMs in the project.*)
 - **Source filter:** `IPv4 Ranges`
 - **Source IP Ranges:** `0.0.0.0/0` (*Allows global access, which is needed for HTTP and SSH access.*)
 - **Protocols and Ports:**
 - ✓ **Allow TCP: 80** (For HTTP traffic)
 - ✓ **Allow TCP: 22** (For SSH remote access)
5. Click **Create** to apply the firewall rule.

allow-http-ssh	
Logs 	
Off	
view in Logs Explorer	
Network	
default	
Priority	
1000	
Direction	
Ingress	
Action on match	
Allow	
Tags	
— 	
Source filters	
IP ranges	0.0.0.0/0
Protocols and ports	
tcp:22, 80	

Testing Firewall Rules

To validate that the firewall rule is working as expected:

I accessed the VM via SSH and executed the following commands:

bash:

```
sudo apt update

sudo apt install -y apache2

sudo systemctl start apache2

sudo systemctl enable apache2
```

Explanation of Commands:

1. `sudo apt update` → Updates package lists for the latest software versions.
2. `sudo apt install -y apache2` → Installs Apache2 web server.
3. `sudo systemctl start apache2` → Starts the Apache2 service.
4. `sudo systemctl enable apache2` → Ensures Apache2 starts automatically on reboot.

For HTTP access:

- I ran `curl http://<EXTERNAL_IP>` from my local machine.
- The expected response (Auto-Scaling VM Running) was received, confirming HTTP access.

```
PS C:\Users\Sujay> curl http://34.135.198.134

StatusCode      : 200
StatusDescription : OK
Content         :
                  <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
                  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
                  <html xmlns="http://www.w3.org/1999/xhtml">
                  <head>
                  <meta http-equiv="Co...
RawContent      : HTTP/1.1 200 OK
                  Vary: Accept-Encoding
                  Keep-Alive: timeout=5, max=100
                  Connection: Keep-Alive
                  Accept-Ranges: bytes
                  Content-Length: 10701
                  Content-Type: text/html
                  Date: Sun, 02 Mar 2025 11:07:54 GM...
Forms           : {}
Headers         : {[Vary, Accept-Encoding], [Keep-Alive, timeout=5, max=100], [Connection, Keep-Alive],
                  [Accept-Ranges, bytes]...}
Images          : {@{innerHTML=; innerText=; outerHTML=<IMG class=floating_element alt="Debian Logo"
                  src="/icons/openlogo-75.png">; outerText=; tagName=IMG; class=floating_element; alt=Debian Logo;
                  src=/icons/openlogo-75.png}}
InputFields     : {}
Links           : {@{innerHTML=manual; innerText=manual; outerHTML=<A href="/manual">manual</A>; outerText=manual;
                  tagName=A; href=/manual}, @{innerHTML=public_html; innerText=public_html; outerHTML=<A
                  href="http://httpd.apache.org/docs/2.4/mod/mod_userdir.html" rel=nofollow>public_html</A>;
                  outerText=public_html; tagName=A; href=http://httpd.apache.org/docs/2.4/mod/mod_userdir.html;
                  rel=nofollow}, @{innerHTML=existing bug reports; innerText=existing bug reports; outerHTML=<A href=
                  "http://bugs.debian.org/cgi-bin/pkgreport.cgi?ordering=normal;archive=0;src=apache2;repeatmerged=0"
                  rel=nofollow>existing bug reports</A>; outerText=existing bug reports; tagName=A; href=http://bugs
                  .debian.org/cgi-bin/pkgreport.cgi?ordering=normal;archive=0;src=apache2;repeatmerged=0;
                  rel=nofollow}}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 10701
```

For SSH access:

- I successfully connected using `ssh b22cs063@<EXTERNAL_IP>`, verifying SSH access was functional.

Architecture Diagram:

